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ENTERPRISE ORCHESTRATOR

ZTF-Orchestrator

ZTF Orchestrator Operational Quick Guides



AHV Appliance Deployment Quick Guide

ZTF-Orchestrator | v1.8.5-rc1

Deploy an approved ZTF-Orchestrator QCOW2 on an existing Nutanix AHV cluster, assign a stable management address, and verify first access.

Documentation baseline: merged source bc79371. Confirm your installed build and local change requirements. Address and hostname examples are reserved for documentation.

How to use this guide

Read all four sections before making changes. Gather the approved network values first. Use the DHCP reservation or static-address path that matches your site, then retain the verification evidence.

Section	Purpose
1 Prepare and deploy the appliance	Verify the QCOW2, import it into Prism and provision SSH access before first boot.
2 Configure the management IP	Set a DHCP reservation or configure static IPv4 with Netplan.
3 Verify first boot and sign in	Check the guest network, service health and installed build.
4 Configure integrations and hand over	Test the required connections and record the operating baseline.

1 Prepare and deploy the appliance

1. Obtain the approved QCOW2 and its matching SHA-256 checksum file. For an air-gapped site, select the airgap build and confirm the application and PostgreSQL images were preloaded.

Prepare	Record or verify
Target	AHV cluster, storage container and VM name.
Guest sizing	Starting point: 2 vCPUs, 4-8 GB RAM and 80-100 GB disk capacity.
Management network	Subnet/VLAN, address assignment method, IP/prefix, gateway, DNS and time source.
Guest access	Approved SSH key, Linux account name, and a working Prism console recovery path.

2. From the extracted artifact directory, verify the checksum. Substitute the filename supplied with your build. Continue when the QCOW2 reports **OK**.

```
sha256sum -c 'SHA256SUMS-airgap-REPLACE_WITH_BUILD_REF.txt'
```

3. Upload the QCOW2 as a **Disk** image under **Infrastructure > Compute & Storage > Images** in Prism Central or **Settings > Image Configuration** in Prism Element where available. Wait until the image is available to the target cluster.

4. Create the VM, clone its boot disk from the uploaded image and put that disk first in the boot order. Attach a virtual NIC to the approved management subnet.

5. Leave the VM powered off while you prepare the SSH key and guest customization on the next page.

Provision SSH access before first boot

1. **Generate a key on your Windows workstation.** Open Command Prompt. Use an unused filename and set a passphrase when prompted. If your site issued a key, use its approved public key instead.

```
if not exist "%USERPROFILE%\.ssh" mkdir "%USERPROFILE%\.ssh"
ssh-keygen -t ed25519 -f "%USERPROFILE%\.ssh\ztf_appliance"
type "%USERPROFILE%\.ssh\ztf_appliance.pub"
```

Copy the entire line from the **.pub** file. Keep the private file **ztf_appliance** on your workstation.

2. **Create ztfadmin before power-on.** In the approved AHV guest customization or cloud-init user-data field, enter the YAML below. Replace the complete example key line with your copied line. Keep the spacing shown. If your site restricts passwordless sudo, use its approved guest-access profile.

```
#cloud-config
users:
  - name: ztfadmin
    groups: sudo
    shell: /bin/bash
    sudo: ALL=(ALL) NOPASSWD:ALL
    ssh_authorized_keys:
      - ssh-ed25519 REPLACE_WITH_YOUR_PUBLIC_KEY operator
ssh_pwauth: false
```

Static-only network: before first boot, ask your provisioning administrator to configure static networking through the approved guest customization method, or arrange a Linux administrator login for the Prism VM console. The SSH key example provides key-based access. For a console password, use the approved console account or a site-managed password hash. Keep SSH password authentication disabled.

Save the customization, then power on. Continue to the management IP page before trying SSH.

2 Configure the management IP

Set the management IP before SSH. If static networking was provisioned, verify the address in Prism or from the VM console. If it was not provisioned, use the Prism VM console with the Linux administrator login arranged before boot. If you have neither an IP nor a console login, ask the provisioning administrator to supply access.

1. **Find the network adapter.** Run these commands on Ubuntu. Find the adapter whose MAC address matches the virtual NIC in Prism. Write down its name, for example **ens3**.

```
ip -br link
ip -br address
ip route
```

2. **Choose one method.** For a DHCP reservation, have the DHCP/IPAM administrator reserve the address for this VM. Leave Ubuntu on DHCP, confirm the address appears, and go to the SSH verification page. For a console-configured static IP, continue below.

3. **Find and back up the file.** Run **sudo ls -l /etc/netplan** and **sudo netplan get**. Identify the existing YAML that configures your adapter. If cloud-init owns it or you cannot identify the right file, ask the provisioning administrator to set the IP at its source. The commands below assume a site-managed file already exists. Replace its example name first.

```
NETPLAN_FILE=/etc/netplan/01-management.yaml
BACKUP="/root/netplan-before-$(date +%Y%m%d-%H%M%S).yaml"
sudo cp -a -- "$NETPLAN_FILE" "$BACKUP" && sudoedit "$NETPLAN_FILE"
```

4. **Edit only the matching adapter.** Replace the example adapter, IP/prefix, gateway and DNS with your approved values. Keep other settings and indent with spaces. In Nano, use Ctrl+O, Enter, then Ctrl+X.

```
network:
  version: 2
  ethernets:
    ens3:
      dhcp4: false
      addresses: [192.0.2.50/24]
      routes:
        - to: default
          via: 192.0.2.1
      nameservers:
        addresses: [192.0.2.53]
```

5. **Restrict file access and start the trial.** Use the same console. If a warning names a different YAML file, check that exact file.

```
sudo chown root:root "$NETPLAN_FILE"
sudo chmod 600 "$NETPLAN_FILE"
sudo netplan generate && sudo netplan try --timeout 180
```

Keep the console open. Check the address and route, then use the SSH verification page before accepting the trial. If the checks fail, let the trial expire and restore the backup from the console.

```
sudo cp -a -- "$BACKUP" "$NETPLAN_FILE" &&
  sudo netplan generate && sudo netplan apply
ip -br address
```

Verify SSH and sudo

Connect from Windows. Use the management IP confirmed on the previous page. Replace **192.0.2.50** with that IP. On first connection, verify the host fingerprint against a trusted deployment record or with your AHV administrator before accepting it.

```
ssh -o IdentitiesOnly=yes -i "%USERPROFILE%\.ssh\ztf_appliance" ^
  ztfadmin@192.0.2.50
```

```
whoami
sudo -v
```

Run these at the Linux prompt. Expect **ztfadmin** and a successful sudo check. The example profile returns silently. If SSH says **Permission denied (publickey)**, check the IP, username and matching key, then ask the AHV administrator to check guest customization.

3 Verify first boot and sign in

1. **Check that the appliance is ready.** Run these commands from Ubuntu. The Orchestrator and PostgreSQL containers should be running. Expect HTTP 200 and healthy from the local check.

```
sudo docker ps --format 'table {{.Names}} {{.Status}}'
curl -fsS http://127.0.0.1:5001/health
```

If either check fails, inspect the first-boot log with **sudo journalctl -u ztf-orchestrator-firstboot -n 100 --no-pager**.

2. **Find the application admin password.** After SSH and sudo succeed, run this in the appliance shell. It shows the first-run banner and the two lines under it.

```
sudo journalctl -u ztf-orchestrator --no-pager |
  grep -F -A 2 'default admin account created'
```

If the journal has no match, check the application container log.

```
sudo docker logs ztf-orchestrator 2>&1 |
  grep -A 2 -F 'default admin account created'
```

Look for **Username: admin** and the generated value after **Password:**. This is the web application password. The Linux account is the one you used to run the command. Keep the generated password out of screenshots, chats and tickets.

If both logs lack the banner: confirm the app is healthy. Use the current web credential if available, or follow the reset on the next page. Restarting an initialized app leaves its saved password in place.

3. **Open the web console.** From a management workstation, open your approved appliance URL. The example direct listener is http://192.0.2.50:5001. Use the trusted HTTPS endpoint for shared access. Sign in as **admin** and rotate the initial password under **Administration > Users**.

4. In **Settings > About > Installed Build**, confirm the intended build. Check the guest route, DNS and time using the commands below. Replace the example hostname with a known internal name.

```
ip route
resolvectl query ztf-orchestrator.example.test
timedatectl status
```

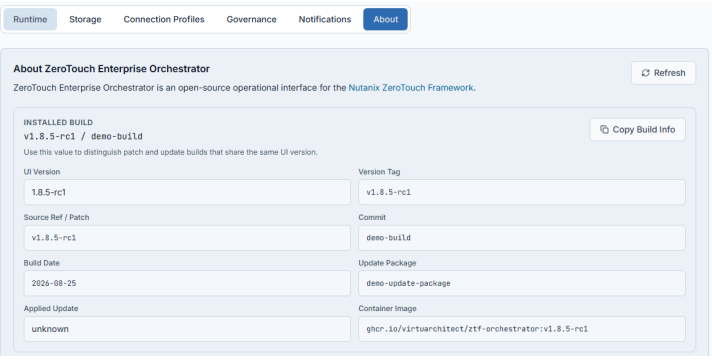


Illustration. Installed Build in the simulated console.

Reset the web admin password

Use this fallback when the appliance is healthy and the web password is unavailable. Run the whole block from an interactive Linux console or SSH session with sudo. Enter a new password twice when prompted.

```
sudo docker exec -it -w /app ztf-orchestrator python -c '
import os
from datetime import datetime, timezone
from getpass import getpass
from pathlib import Path

import bcrypt
from storage_backend import create_storage

data_dir = Path(os.environ.get("ZTF_DATA_DIR", str(Path.home() / ".ztf-ui")))
storage = create_storage(data_dir)
users_path = data_dir / "users.json"
users = storage.read_json(users_path, [])
matches = [user for user in users if user.get("username") == "admin"]

if len(matches) != 1 or matches[0].get("role") != "admin":
    raise SystemExit("Stopped: expected one existing admin account with the admin role.")

password = getpass("New ZTF web admin password: ")
confirmation = getpass("Confirm new password: ")
if password != confirmation:
    raise SystemExit("Stopped: passwords differ.")
if len(password) < 12 or len(password.encode("utf-8")) > 72:
    raise SystemExit("Stopped: use at least 12 characters and at most 72 UTF-8 bytes.")

matches[0]["password_hash"] = bcrypt.hashpw(
    password.encode("utf-8"), bcrypt.gensalt()
).decode("ascii")
matches[0]["password_changed_at"] = datetime.now(timezone.utc).isoformat()
storage.write_json(users_path, users)
print("ZTF web admin password reset successfully.")
'
```

Expect the success message. In a private browser window, sign in as admin with the new password. If the script reports that it cannot find one existing admin account with the admin role, keep the user store intact. Ask another administrator to review the account, or send the appliance maintainer the installed build and redacted startup errors.

4 Configure integrations and hand over

1. Add the Prism or Foundation connections and credential references your workflows require. Verify certificate trust and access from the appliance.

2. Confirm that an authorized administrator can sign in over the management path. Keep credentials in the approved store and retain these details in the handover record.

Appliance record	Retain independently
Image and VM	Approved QCOW2 build/checksum, AHV cluster, VM name and disk capacity.
Network	Management subnet, IP/prefix, gateway, DNS, and DHCP reservation or static assignment.
Health	Local health result, time synchronization and management workstation reachability.
Access	Approved HTTPS address, Linux account owner and verified SSH/sudo access.
Build	Installed version, source revision and image tag shown under Installed Build.
Integrations	Connection tests, credential reference names and unresolved exceptions.

If local health passes but workstation access fails, check the VM subnet, network routes, firewall policy and configured listener. Verify the trusted HTTPS endpoint separately where applicable.

Expected: operators can reach the appliance at its approved management address, sign in and identify the installed build.

Source and review boundary

Procedure checked against the pinned repository source and platform references. The screenshot uses generic demo data. Site validation is performed on the deployed appliance.

ZTF-Orchestrator appliance kit

Nutanix Prism Element Web Console Guide

Netplan trial and rollback